Imperial College
Business School

Imperial means
Intelligent Business

Professional Certificate in ML&AI

Office Hour-10

13-May-26, 8pm UTC

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📋 Agenda

- Random Forest Hyperparameters

Random Forest Model- Tuning

RandomForestClassifier

```
class sklearn.ensemble.RandomForestClassifier(n_estimators=100, *,
criterion='gini', max_depth=None, min_samples_split=2, min_samples_leaf=1,
min_weight_fraction_leaf=0.0, max_features='sqrt', max_leaf_nodes=None,
min_impurity_decrease=0.0, bootstrap=True, oob_score=False, n_jobs=None,
random_state=None, verbose=0, warm_start=False, class_weight=None, ccp_alpha=0.0,
max_samples=None, monotonic_cst=None)
```

[source]

n_estimators = no of trees

- Default is 100
- You can extend it to 200/ 300
- Go to 500 only if there is an improvement

More Trees- usually better and stable predictions
Too many trees - slower training

max_depth = height of each tree

- The most important parameter for controlling overfitting
- Try: None, 3, 5, 7, 10 and so on

Small depth ➔ simple model ➔ less overfitting

Large depth ➔ complex model ➔ overfitting

So if :

- overfitting ➔ reduce depth
- underfitting ➔ increase depth

max_features = how many features each tree sees

- Common ones: sqrt, log2, None
- Lower values → fewer features→ trees more different
- higher values → more features → trees more similar

Helps reduce overfitting as trees can become less or more correlated

min_samples_leaf = min samples allowed in a leaf

- Useful for controlling overfitting

- 1 ➔ very flexible trees (model memorizes tiny details)
- higher values ➔ smoother predictions (model learns broad patterns)

min_samples_split = min samples needed before splitting a node

- higher values ➔ harder to create splits➔ simpler trees
- not as impactful as max_depth and min_samples_leaf

What is controlled?

- max_depth ➔ controls intelligence/complexity
- min_samples_leaf ➔ controls smoothness
- max_features ➔ controls diversity between trees
- n_estimators ➔ controls stability

Your Feedback is important!

📝Feedback

🔗**Please provide your valuable feedback on the Module and OH in the Canvas platform**


Q
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